minimum or saddle point) of the Action ***S***. The conditions that produce an extremum are a set of "Euler-Lagrange" equations that control the evolutions of the fields. This is a standard procedure in the calculus of variations. The Professor and Murph make guesses for a list of unknown bulk fields φ^i and unknown functions $U(Q)$, $H_{ij}(Q^2)$, $\mathcal{M}$(standard model fields), and unknown constants W_{ij} that appear in the Lagrangian. In Figure 25.9 you see me writing a list of their guesses on the blackboard. Then for each set of guesses, they vary the fields and spacetime geometries, deduce the Euler-Lagrange equations, and then explore in computer simulations those equations' predictions for the gravitational anomalies.

Chapter 27. The Volcano's Rim

This note is for readers who are familiar with the mathematical description of Newton's laws of gravity and the conservation of energy and angular momentum. I challenge you to deduce the following formula for the volcano-like surface from (i) the Paczynski-Wiita approximate formula for Gargantua's gravitational acceleration, $g = GM/(r - r_h)^2$ (see the technical notes for Chapter 17, above) and (ii) the conservation laws for energy and angular momentum. The formula, using the notation of the technical notes for Chapter 17 plus L for the *Endurance*'s angular momentum (per unit mass), is

$$V(r) = -\frac{GM}{r - r_h} + \frac{1}{2}\frac{L^2}{r^2}.$$

The first term is the *Endurance*'s gravitational energy (per unit mass), the second is its circumferential kinetic energy, and the sum of $V(r)$ and the radial kinetic energy $v^2/2$ (with v its radial velocity) is equal to the *Endurance*'s conserved total energy (per unit mass). The rim of the volcano is at the radius r where $V(r)$ is a maximum. I challenge you,